

R1

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1 New Code / Devs

1.1 Python Terminal Script for Filtering

Due to the fact that we to filter multiple simulations with very similar methods, I decided to make an actual script instead of just using a notebook repeatably. The largest upside is that this can easily be run on a compute node, which makes filtering larger sims easier. Also helps make sure any results are reproducible.

Takes a user input via the CLI (see below), filters for all **BH_Sol** systems, and returns the population file with said systems. Additionally, it returns a .csv of just the **BH_Sol** rows. While running it provides feedback via the terminal stating how many **BH_MS** systems it found, as well as how many are the expected **S1:S2** configuration. If any are **S2:S1**, it prints a warning as well as the total count.

Need to further refine it, but there's also a default SBATCH script to help expedite the submission process.

Usage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
-o	output location	.results
-outputName	name for the outputed files ¹	BH_Sol_Systems
-overwrite	Whether to overwrite existing results, T/F bool. Defaults to F	True
--maxMass	Max mass used when filtering, defaults to $3M_{\odot}$	3.5
--maxPeriod	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

¹If a name is not specified it will use the original filename, appending a specifier to the front

2 New Research

2.1 Pop 1

When filtering via notebook and script, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, ZAMS_oCE1_CC1_oRL02_CC2. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then into post-MS **unstable** RLO via $S1 \rightarrow S2$. This period decrease leads to common envelope, which then *greatly* decreases the orbital period. The envelope is then ejected, leaving a depleted stripped He core². The core then collapses in a couple of thousand years, leaving the system in a BH-MS state with low p_{orb} /seperation.

These systems then evolve in a semi-peculiar way, where it is highly likely that there will be RLO from S2 onto S1, and because $S1 > S2$, the transfer is stable, leading to *very* prolonged stable RLO (seemingly on the scale of $> 10Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestingly, the initial period for all the systems is close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the initial masses of S1 and S2 appear to be quite constrained as well.

	orbital_period.i		S1_mass.i		S2_mass.i
count	6.000000	count	6.000000	count	6.000000
mean	4022.716197	mean	17.515761	mean	2.296788
std	151.522384	std	0.520269	std	0.320702
min	3873.118288	min	16.815111	min	1.797809
25%	3908.300983	25%	17.190036	25%	2.120753
50%	3978.305326	50%	17.510176	50%	2.353828
75%	4124.397101	75%	17.818155	75%	2.547210
max	4248.507680	max	18.254965	max	2.620529

2.2 Pop 2

Ran a second grid including an eccentricity distro (Ps and Qs), as well as slightly constraining the dataset further ($12 < M_1 < 100, .1 < M_2 < 10$)³, hoping to confirm some of the initial conditions.

	orbital_period.i		S1_mass.i		S2_mass.i
count	12.000000	count	12.000000	count	12.000000
mean	4136.066721	mean	2.240382	mean	17.357478
std	177.676729	std	0.412075	std	1.073496
min	3844.570082	min	1.551266	min	15.506937
25%	4024.214908	25%	1.903135	25%	16.714083
50%	4119.397535	50%	2.300024	50%	16.998229
75%	4273.345845	75%	2.579111	75%	18.162633
max	4370.857404	max	2.729828	max	19.435080

²I need to read a lot more about CE evo, but my intuition feels weird about the fact that just the envelope of *one* of the stars is ejected

³This is very constrained very early, but I wanted to try something while waiting for the 10 million sim

2.3 Full P_{orb}

When filtering pop 2 with a max $p_{orb} = 300$ days we find 46 BH-Sol systems, all of these (except one!) evolve via the same CE channel. There is one system which does *not* do RLO before CC1, but only reaches a solar-like mass after RLO2 (and also fails), so I'm excluding it for now. Probably important to keep in mind though.

When plotting S1 vs S2 initial and formation masses there appears to be a correlation. In fig 2 this correlation appears to be quite strong, however, this very well may be because of the initial distribution.

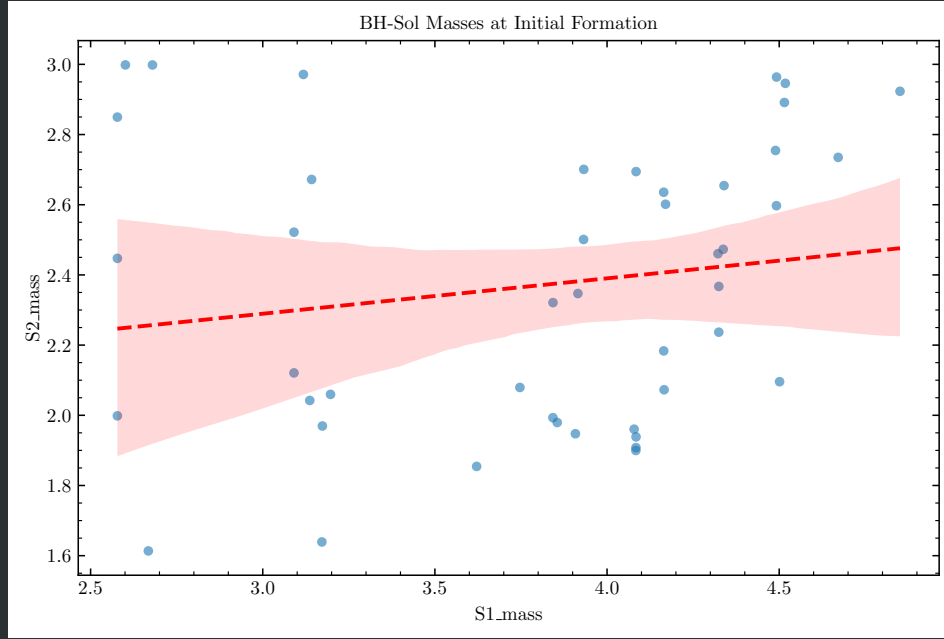


Figure 1: Masses at initial BH-Sol state. i.e. BH formation/S1 collapse

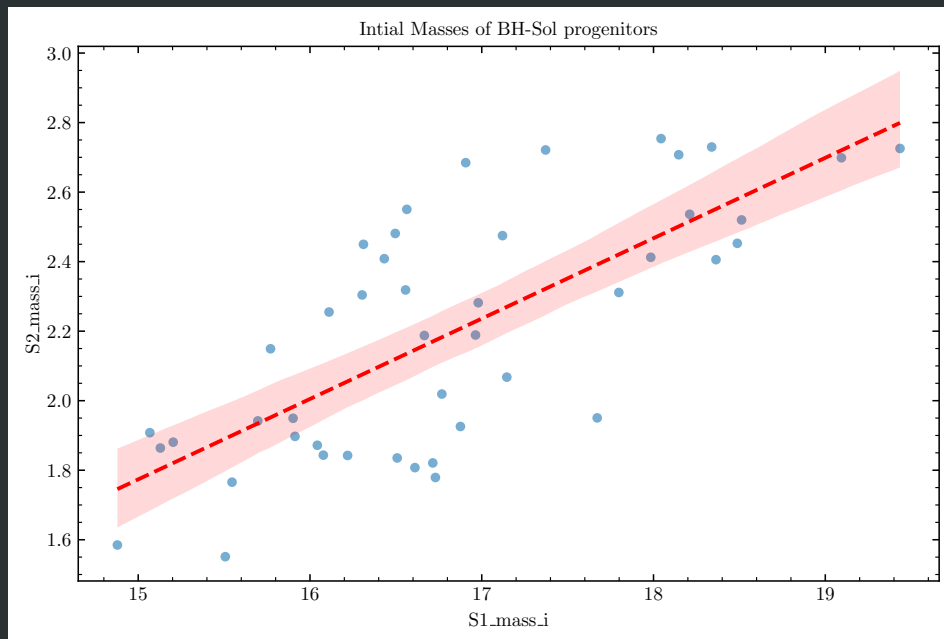


Figure 2: Initial ZAMS masses. May just be a byproduct of Ps and Qs, further sims with diff initial distribution needed.

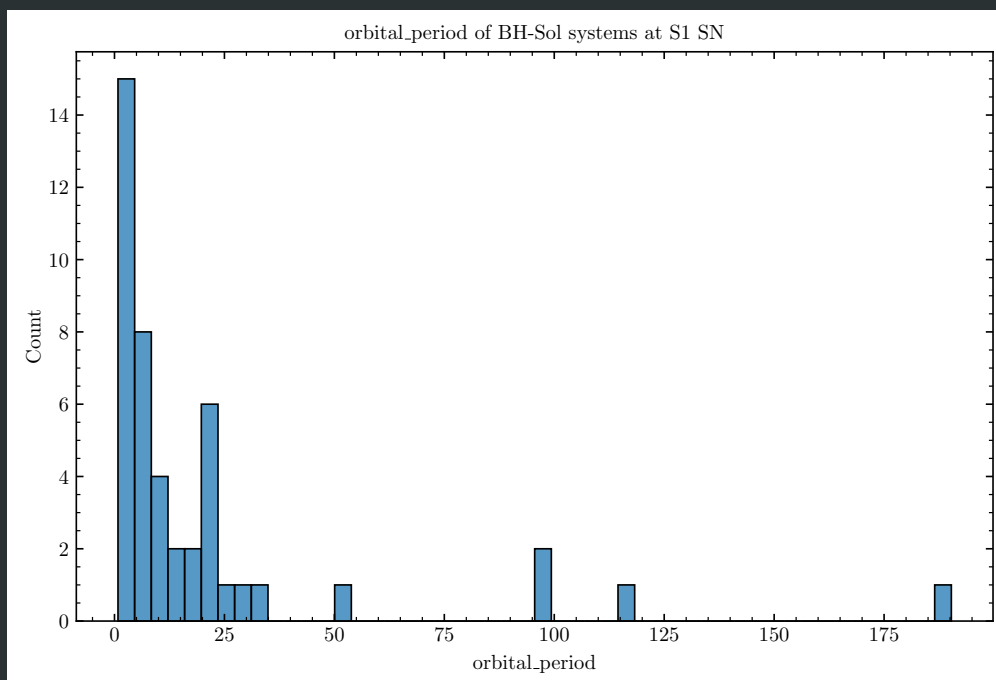


Figure 3: Orbital Period of BH-Sol Systems at init onset.

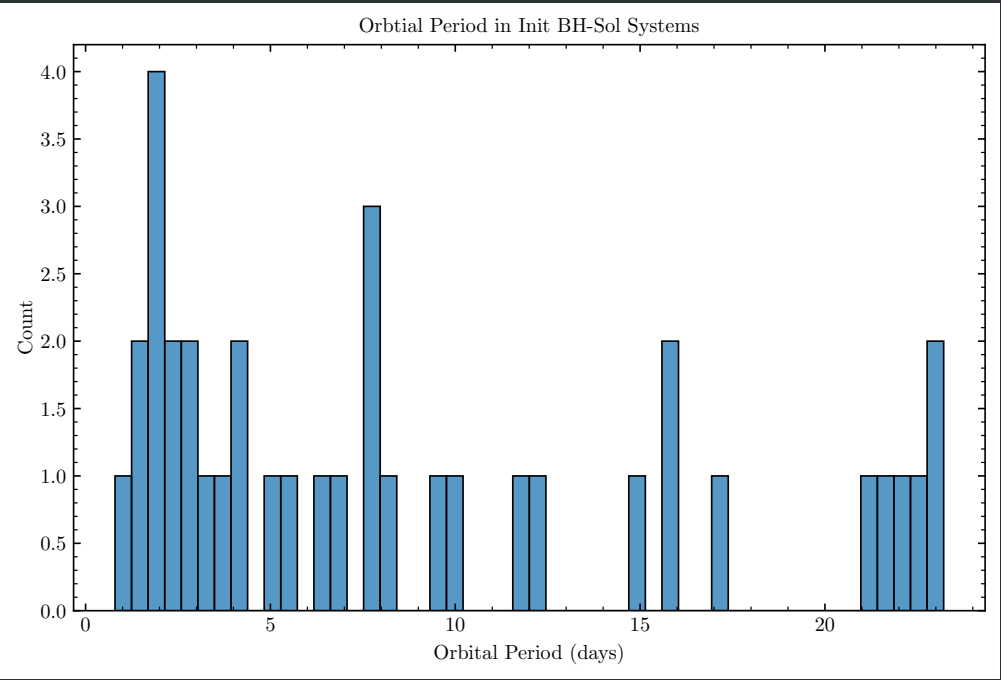


Figure 4: Crop of orbital Period of BH-Sol Systems at init onsent.

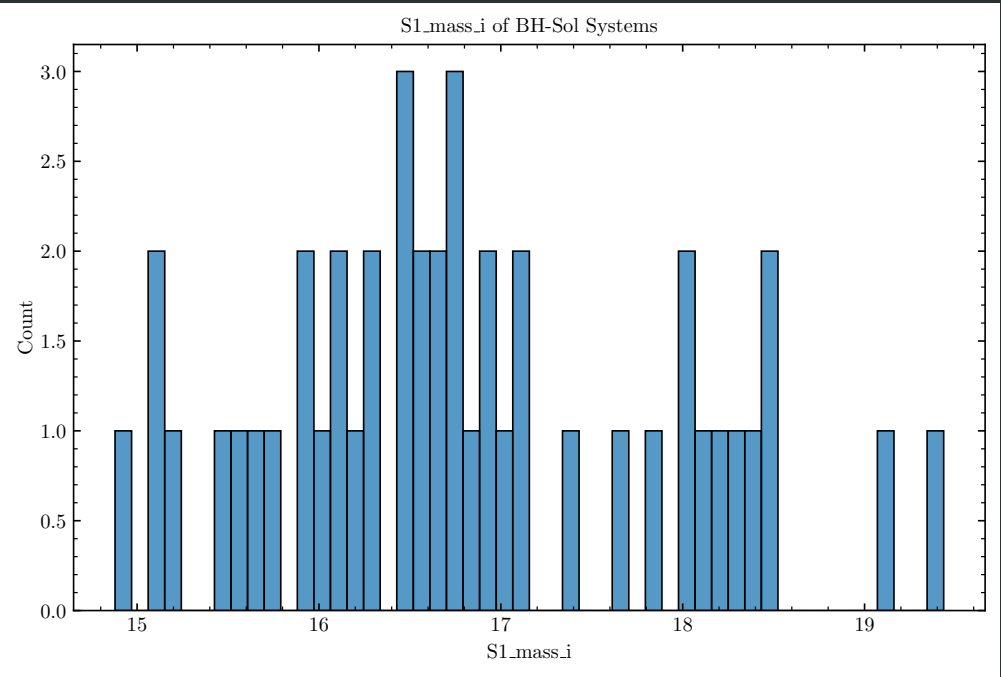


Figure 5: S1 mass initial distribution

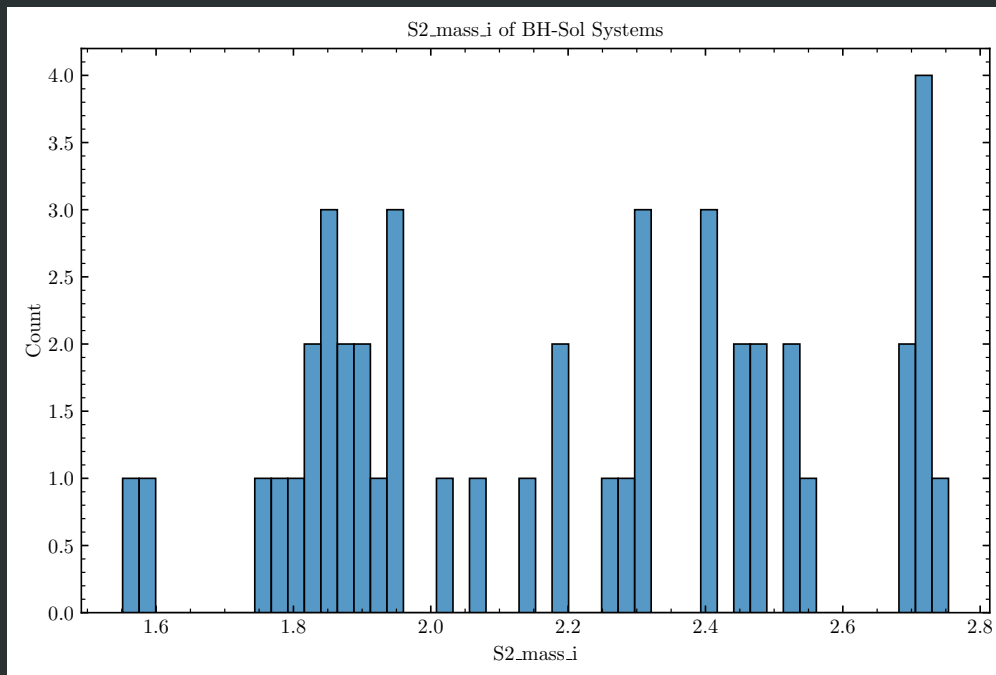


Figure 6: S2 mass initial distribution

3 To-Do

- Find out if it makes sense for the onset of un-stable rlo to happen at $P_{orb} \approx 4000\text{days}$

4 Questions

- Tried running a grid, seems to have just completely failed from trying to reference itself before it was fully initialized. Entire log was filled with

```
Traceback (most recent call last):
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/popsyn/binarypopulation.py", 1
    binary.evolve()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
    self.run_step()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
    next_step(self)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_merged.py", 1
    super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_isolated.py", 1
    super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.py", 1
    self.update_after_evo(t, binary, primary, secondary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.py", 1
    current = history[-1]
                ~~~~~~
IndexError: index -1 is out of bounds for axis 0 with size 0
```

and from what I can tell, this happened to nearly all the simmed binaries

- Should we include an initial eccentricity distribution? I imagine it would impact formation with unstable RLO and CE.
- Does it make sense to still be filtering for $p_{orb} < 3\text{days}$? The simulations seem to be suggesting that *all* BH-Sol systems evolve through CE, and that the main thing effecting p_{orb} post CE is the kick strength of the BH. So, maybe we should be initially filtering for all BH-Sol, and if we want the formation rate of specifically low p_{orb} ones, just take the subpop.
- Cool if I present an assortment of general POSYDON things I've worked on at astrofest?